

[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1427

C

Unique Paper Code : 32351303

Name of the Paper : BMATH 307 – Multivariate
Calculus

Name of the Course : B.Sc. (H) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All sections are compulsory.
3. Attempt any Five questions from each section. All questions carry equal marks

SECTION I

1. Let $f(x, y) = \frac{xy(x^2 - y^2)x}{x^2 + y^2}$ if $(x, y) \neq (0, 0)$
= 0 otherwise

Show that $f(0, y) = -y$ and $f(x, 0) = x$ for all x and y .

P.T.O.

$$\vec{F}(x, y) = (3y - 4x)\hat{i} + (4x - y)\hat{j}$$

when an object moves once counterclockwise around the ellipse $4x^2 + y^2 = 4$.

4. Use Stoke's theorem to evaluate the surface integral

$$\iint_S (\text{curl } \vec{F} \cdot \vec{N}) dS$$

where $F = x\hat{i} + y^2\hat{j} + ze^{xy}\hat{k}$ and S is that part of surface $z = 1 - x^2 - 2y^2$ with $z \geq 0$.

5. Use divergence theorem to evaluate the integral

$$\iint_S \vec{F} \cdot \vec{N} dS \quad \text{where} \quad \vec{F}(x, y, z) = (\cos yz)\hat{i} + e^{xz}\hat{j} + 3z^2\hat{k},$$

where S is hemisphere surface $z = \sqrt{4 - x^2 - y^2}$ together with the disk $x^2 + y^2 \leq 4$, in x - y plane.

6. Evaluate the line integral $\int_C \vec{F} \cdot d\vec{R}$

Where $\vec{F}(x, y) = [(2x - x^2y)e^{-xy} + \tan^{-1}y]\hat{i} +$

$\left[\frac{x}{y^2 + 1} - x^3 e^{-xy} \right]\hat{j}$ and C is the ellipse $9x^2 + 4y^2 = 36$.

Name of the Course	: B.Sc. (Hons.) Mathematics CBCS (LOCF)
Unique Paper Code	: 32351303
Name of the Paper	: BMATH307 – Multivariate Calculus
Semester	: III
Duration	: 3 Hours
Maximum Marks	: 75

Attempt any four questions. All questions carry equal marks.

1. Let $f(x, y) = \begin{cases} \frac{y^3}{x^2+y^2}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$

Is the function f continuous at $(0,0)$? Justify your answer.

Find an equation for the tangent plane to the surface $z = f(x, y)$ defined above at the point $P_0\left(1, 2, \frac{8}{5}\right)$.

Also find the directional derivative of $f(x, y)$ at $P_0(1,2)$ in the direction of $\mathbf{v} = 3\mathbf{i} - 2\mathbf{j}$.

2. Find the critical point and classify each point as a relative minimum, relative maximum, or a saddle point of

$$f(x, y) = xye^{-8(x^2+y^2)}.$$

Find the maximum and minimum values of $f(x, y, z) = xyz$ subject to the constraint

$$x^2 + 2y^2 + 4z^2 = 24.$$

Where is the function $f(x, y) = \sqrt{x^2 + y^2}$ differentiable?

3. Compute $\iint_R xe^{xy} dA$ where R is the rectangle $0 \leq x \leq 1, 1 \leq y \leq 2$, using iterated integrals in both orders.

Evaluate $\iint_R 6x^2y dA$ if R is the region bounded between the curves $y = x, y = 1$ and $4y = x^2$.

Find the area of the region bounded between the curves $r_1(\theta) = 2 + \sin 3\theta$ and $r_2(\theta) = 4 - \cos 3\theta$.

4. Find the mass of the ellipsoid $4x^2 + 4y^2 + z^2 = 16$ lying above the xy -plane if the density is given by $\delta(x, y, z) = z$.

Determine the centroid of the solid bounded above by the sphere $x^2 + y^2 + z^2 = 1$ and below by the xy -plane where the density is given by $\delta(x, y, z) = z$.

Compute $\int_0^1 \int_0^1 x^2 y \, dx \, dy$ by changing $u = x$ and $v = xy$.

5. Evaluate $\oint_C (x^2 z dx - y x^2 dy + 3 dz)$ where C is the boundary of the triangle with vertices $(0, 0, 0)$, $(1, 1, 0)$ and $(1, 1, 1)$.

Find a non-zero function h for which

$$\mathbf{F}(x, y) = h(x)(x \sin y + y \cos y)\mathbf{i} + h(x)(x \cos y - y \sin y)\mathbf{j}$$

is conservative.

Using line integral, find the area of the region enclosed by the asteroid

$$x = a \cos^3 t, \quad y = a \sin^3 t \quad (0 \leq t \leq 2\pi).$$

6. Find the mass of the lamina that is the portion of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 2$ with constant density δ_0 .

Verify Stokes' Theorem if $\mathbf{F}(x, y, z) = (x - y)\mathbf{i} + (y - z)\mathbf{j} + (z - x)\mathbf{k}$ and S be the portion of the plane $x + y + z = 1$ in the first octant assuming that the surface has an upward orientation.

Using the Divergence Theorem, evaluate $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = (z^3)\mathbf{i} - x^3\mathbf{j} + y^3\mathbf{k}$ and S is the sphere $x^2 + y^2 + z^2 = a^2$, with outward unit normal vector $\mathbf{N}$.

[This question paper contains 7 printed pages]

Your Roll No. :

Sl. No. of Q. Paper : 7465 J

Unique Paper Code : 32351303

Name of the Course : B.Sc.(Hons.)
Mathematics

Name of the Paper : Multivariate Calculus

Semester : III

Time : 3 Hours

Maximum Marks : 75

Instructions for Candidates :

- (i) Write your Roll No. on the top immediately on receipt of this question paper.
- (ii) **All** Sections are compulsory.
- (iii) Attempt any **five** questions from each **Section**.
- (iv) All questions carry equal marks.

Section- I

1. Given that the function

$$f(x, y) = \begin{cases} \frac{3x^3 - 3y^3}{x^2 - y^2} & \text{for } x^2 \neq y^2 \\ B & \text{otherwise} \end{cases}$$

is continuous at the origin, what is B ?

2. In physics, the wave equation is :

$$\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$$

and the heat equation is :

$$\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2}$$

Determine whether $z = \sin 5ct \cos 5x$ satisfies the wave equation, the heat equation, or neither.

3. The radius and height of a right circular cone are measured with errors of at most 3% and 2%, respectively. Use increments to approximate the maximum possible percentage error in computing the volume of the cone using these measurements and the formula $V = \frac{1}{3}\pi R^2 H$.

4. If $f(x, y, z) = xy^2e^{xz}$ and $x = 2 + 3t$, $y = 6 - 4t$, $z = t^2$. Compute $\frac{df}{dt}(1)$.

5. Sketch the level curve corresponding to $C = 1$ for the function $f(x, y) = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ and find a unit normal vector at the point $P_0(2\sqrt{3})$.

6. Find the point on the plane $2x + y - z = 5$ that is closest to the origin.

Section - II

7. Find the volume of the solid bounded above by the plane $z = y$ and below in the xy -plane by the part of the disk $x^2 + y^2 \leq 1$ in the first quadrant.
8. Sketch the region of integration and then compute the integral $\int_0^1 \int_x^{2x} e^{y-x} dy dx$ in 2 ways:
(a) with the given order of integration
(b) with the order of integration reversed
9. Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} y \sqrt{x^2 + y^2} dy dx$ by converting to polar coordinates.
10. Find the volume of the tetrahedron bounded by the plane $2x + y + 3z = 6$ and the coordinate planes $x = 0$, $y = 0$ and $z = 0$.

11. Compute $\iiint_D \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ where D is the solid

sphere $x^2 + y^2 + z^2 \leq 3$.

12. Use the change of variables to compute

$\iint_D \frac{(x-y)^4}{(x+y)^4} dy dx$, where D is the triangular

region bounded by the line $x + y = 1$ and the coordinate axes.

Section - III

13. Find the work done by the force field

$$\vec{F} = \frac{x}{\sqrt{x^2 + y^2}} \vec{i} - \frac{y}{\sqrt{x^2 + y^2}} \vec{j} \text{ when an object moves}$$

from $(a, 0)$ to $(0, a)$ on the path $x^2 + y^2 = a^2$.

14. Verify that the following line integral is

independent of the path $\oint (3x^2 + 2x + y^2) dx + (2xy + y^3) dy$ where C is any path from $(0, 0)$ to $(0, 1)$.



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15. Use Green's theorem to evaluate $\oint_C (x \sin x dx - \exp(y^2) dy)$ where C is the closed

curve joining the points $(1, -1)$, $(2, 5)$ and $(-1, -1)$ in counterclockwise direction.

16. State Stoke's theorem and use it to evaluate

$\iint_S \text{curl} \vec{F} \cdot d\vec{S}$ where $\vec{F} = xz\vec{i} + yz\vec{j} + xy\vec{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

17. Use the divergence theorem to evaluate the

surface integral $\iint_S \vec{F} \cdot \vec{N} dS$, where $\vec{F} = (x^2 + y^2 - z^2)\vec{i} + yx^2\vec{j} + 3z\vec{k}$; S is the surface comprised of the five faces of the unit cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, missing $z = 0$.

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18. Evaluate $\iint_S 2x \, dS$ where S is the portion of the plane $x + y + z = 1$ with $x \geq 0, y \geq 0, z \geq 0$.



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Sl. No. of Ques. Paper : 1356

F-7

Unique Paper Code : 2351503

Name of Paper : Calculus – II (Multivariate Calculus)

Name of Course : B.Sc. (Hons.) Mathematics (Erstwhile FYUP)

Semester : V

Duration : 3 hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

All Sections are compulsory. Attempt any five questions from each Section.

Section-I

1. (a) Let f be the function defined by :

$$f(x, y) = \begin{cases} \frac{xy^3}{x^2 + y^6}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Is f continuous at $(0, 0)$? Justify your answer.

- (b) Compute the slope of the tangent line to the graph of $f(x, y) = x \ln(x + y^2)$ at the point

$P_0(e, 0, e)$ in the direction parallel to the xz -plane.

$$\left(2\frac{1}{2} + 2\frac{1}{2} \right)$$

2. A closed box is found to have length 2 ft, width 4 ft and height 3 ft where the measurement of each dimension is made with a maximum possible error of ± 0.02 ft. The top of the box is made from material that costs \$ 2/ft² and the material for sides and bottom costs only \$ 1.50/ft². What is the maximum error involved in the computation of cost of the box?

(5)

3. Find $\frac{\partial w}{\partial s}$,

if $w = 4x + y^2 + z^3$, where $x = e^{rs}$, $y = \ln \frac{r+s}{t}$, and $z = rst^2$.

(5)

4. (a) Find the gradient of the function $f(x, y, z) = \frac{xy-1}{z+x}$.

- (b) Find the directional derivative of $f(x, y) = x^2 + xy + y^2$ at the point $P_0(1, -1)$ in the direction towards the origin. $\left(2\frac{1}{2} + 2\frac{1}{2}\right)$

5. Find the absolute extrema of the function $f(x, y) = 2\sin x + 5\cos y$ on the closed bounded rectangular region with vertices $(0, 0)$, $(2, 0)$, $(2, 5)$ and $(0, 5)$ (5)

6. Find the maximum and minimum distance from the origin to the ellipse $5x^2 - 6xy + 5y^2 = 4$ (5)

Section-II

7. Sketch the region of integration and write an equivalent integral with the order of integration reversed:

$$\int_{-1}^2 \int_{x^2-2}^x f(x, y) dy dx \quad (5)$$

8. Evaluate

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} y\sqrt{x^2+y^2} dy dx$$

by converting to polar coordinates.

(5)

9. Use a suitable change of variables to compute

$$\iint_D \exp\left(\frac{y-x}{y+x}\right) dy dx,$$

Where D is the triangular region with vertices $(0, 0)$, $(2, 0)$, $(0, 2)$.

(5)

10. Find the volume of the tetrahedron T bounded by the plane $2x + y + 3z = 6$ and the coordinate planes $x = 0$, $y = 0$, and $z = 0$ (5)

11. Evaluate the integral

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{\sqrt{2-x^2-y^2}} z dz dy dx$$

by transforming to cylindrical coordinates.

(5)

12. Find the average value of the function $f(x, y, z) = x + y + z$ over the sphere

$$x^2 + y^2 + z^2 = 4.$$

(5)

Section-III

13. Evaluate

$$\int_C [5xy \, dx + 10yz \, dy + z \, dz],$$

where C is the parabolic arc $x = y^2$ from $(0, 0, 0)$ to $(1, 1, 0)$ followed by the line segment given by $x = 1, y = 1, 0 \leq z \leq 1$.

(5)

14. Prove that the force field $\vec{F} = [(2x - x^2y)e^{-xy} + \tan^{-1} y] \hat{i} + \left[\frac{x}{y^2 + 1} - x^3 e^{-xy} \right] \hat{j}$ is conservative in $\mathbb{R}^2$ and using this evaluate the line integral $\int_C \vec{F} \cdot d\vec{R}$, where C is curve with parametric equations $x = t^2 \cos \pi t, y = e^{-t} \sin \pi t, 0 \leq t \leq 1$.

(5)

15. State Green's theorem for a simply connected region in $\mathbb{R}^2$. Use it to find the work done by the force field $\vec{F}(x, y) = (3y - 4x) \hat{i} + (4x - y) \hat{j}$ when an object moves once counter-clockwise around the ellipse $4x^2 + y^2 = 4$.

(5)

16. Evaluate

$$\iiint_S (x + y + z) \, dS,$$

where S is the surface defined parametrically by

$$\vec{R}(u, v) = (2u + v) \hat{i} + (u - 2v) \hat{j} + (u + 3v) \hat{k} \text{ for } 0 \leq u \leq 1, 0 \leq v \leq 2.$$

(5)

17. Use Stokes' theorem to evaluate

$$\oint_C [(x+2z)dx + (y-x)dy + (z-y)dz],$$

where C is the boundary of the triangular region with vertices $(3, 0, 0)$, $(0, \frac{3}{2}, 0)$, $(0, 0, 3)$ traversed counter clockwise as viewed from above.

(5)

18. Use the divergence theorem to evaluate the surface integral $\iint_S \vec{F} \cdot \vec{N} dS$, where

$\vec{F}(x, y, z) = \cos yz \hat{i} + e^{xz} \hat{j} + 3z^2 \hat{k}$ and S is the closed hemispherical surface

$z = \sqrt{4 - x^2 - y^2}$ together with the disk $x^2 + y^2 \leq 4$ in the xy -plane and $\vec{N}$ is the outward unit normal vector field.

(5)

19. State and prove the Gauss-divergence theorem.

(5)

[This question paper contains 4 printed pages.]

Sr. No. of Question Paper : 2066

GC-3

Your Roll No.....

Unique Paper Code : 32351303

Name of the Paper : C-7 Multivariate Calculus (Including Practicals)

Name of the Course : B.Sc. (Hons.) Mathematics – II – C.B.C.S.

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on the receipt of this question paper.
2. All sections are compulsory.
3. Attempt any five questions from each section.
4. All questions carry equal marks.

SECTION I

1. Show that f is continuous at $(0, 0)$ where

$$f(x, y) = \begin{cases} y \sin 1/x & x \neq 0 \\ 0 & x = 0 \end{cases}$$

2. A closed box is found to have length 2 ft, width 4 ft and height 3 ft, where the measurement of each dimension is made with a maximum possible error of ± 0.02 ft. The top of the box is made from material that costs \$ 2/ft²; and the material for the sides and bottom costs only \$ 1.50/ft². What is the maximum error involved in the computation of the cost of the box ?

3. If $z = f(u - v, v - u)$, show that

$$\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = 0$$

4. Suppose at point $P_0(-1, 2)$, a certain function $f(x, y)$ has directional derivative 8 in the direction of $\mathbf{v}_1 = 3\mathbf{i} - 4\mathbf{j}$ and 1 in the direction of $\mathbf{v}_2 = 12\mathbf{i} + 5\mathbf{j}$. What is the directional derivative of f at P_0 in the direction of $\mathbf{v} = 3\mathbf{i} - 5\mathbf{j}$?
5. Find the absolute extrema of $f(x, y) = xy - 2x - 5y$ on the closed bounded set S where S is the triangular region with vertices $(0, 0)$, $(7, 0)$ and $(7, 7)$.
6. Find the point of intersection of the plane $x + 2y + z = 10$ and the paraboloid $z = x^2 + y^2$ that is closest to the origin. You may assume that such a point exists.

SECTION II

7. Compute $\iint_R \frac{xy}{x^2 + y^2} dA$, where R is the rectangle $1 \leq x \leq 3$, $1 \leq y \leq 2$.

8. Evaluate the integral $\int_0^1 \int_{\tan^{-1}y}^{\pi/4} \sec x \, dx \, dy$.

9. Evaluate $\iint_D \frac{1}{x} dA$, where D is the region that lies inside the circle $r = 3\cos\theta$ and outside the cardioid $r = 1 + \cos\theta$.
10. Find the volume V of the solid, using spherical coordinates, bounded by the sphere $(x-1)^2 + (y-2)^2 + (z-3)^2 = 1$.

11. Find the volume of the solid D where D is the intersection of the solid sphere $x^2 + y^2 + z^2 \leq 9$ and the solid cylinder $x^2 + y^2 \leq 1$.

12. Evaluate $\iint_D e^{-(4x^2+5y^2)} dA$, using change of variable, where D is the elliptical

disk $\frac{x^2}{5} + \frac{y^2}{4} \leq 1$.

SECTION III

13. Let $F(x, y) = e^y i + xe^y j$. Verify that the vector field F is conservative on the entire xy -plane. Use a potential function of F to find the work done by the field F on a particle that moves in anticlockwise direction from $(1, 0)$ to $(-1, 0)$ along the semi-circular path

$$y = \sqrt{1-x^2}; \quad -1 \leq x \leq 1.$$

14. State Green's theorem for simply connected regions. Verify Green's theorem for the field $F(x, y) = (x - y)i + xj$ and the region R bounded by the unit circle $C: r(t) = \cos t i + \sin t j; 0 \leq t \leq 2\pi$.

15. Let E be the solid bounded by the xy -plane and the paraboloid $z = 4 - x^2 - y^2$. Let S be the boundary of E (that is, piece of the paraboloid and a disk in the xy -plane), oriented with the outward-pointing normal n .

If $F(x, y, z) = (xz \sin(yz) + x^3)i + \cos(yz)j + (3zy^2 - e^{x^2+y^2})k$.

Find $\iint F \cdot n \, ds$ over the surface S using divergence theorem.

16. Evaluate $\iint (x + y + z) \, ds$ over the surface S defined parametrically by

$R(u, v) = (2u + v)i + (u - 2v)j + (u + 3v)k$ for $0 \leq u \leq 1; 0 \leq v \leq 2$.

x y z

17. State and prove the Stokes' theorem.

18. State fundamental theorem of line integrals. If a semi-circular wire has the equation $y = \sqrt{25 - x^2}$ and its mass density function is $\psi(x, y) = 15 - y$. Find the mass of the wire. ✓

This question paper contains 4+1 printed pages]

Roll No.

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S. No. of Question Paper : 6625

Unique Paper Code : 32351303 HC

Name of the Paper : Multivariate Calculus

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper)

All sections are compulsory.

Attempt any five questions from each Section.

All questions carry equal marks.

Section I

1. Let f be the function defined by :

$$f(x, y) = \begin{cases} \frac{xy^3}{x^2 + y^6} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}$$

Is f continuous at $(0, 0)$? Explain.

P.T.O.

2. Find the equation for each horizontal tangent plane to the surface :

$$z = 5 - x^2 - y^2 + 4y.$$

3. Let f and g be twice differentiable functions of one variable and let $u(x, t) = f(x + ct) + g(x - ct)$ for a constant c . Show that :

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

4. Let f have continuous partial derivatives and suppose the maximal directional derivative of f at $P_0(1, 2)$ has magnitude 50 and is attained in the direction from P_0 towards $Q(3, -4)$. Use this information to find $\nabla f(1, 2)$.

5. Find the absolute extrema of $f(x, y) = x^2 + xy + y^2$ on the closed bounded set S where S is the disk $x^2 + y^2 \leq 1$.

6. Find the point on the plane $2x + y + z = 1$ that is nearest to the origin.

Section II

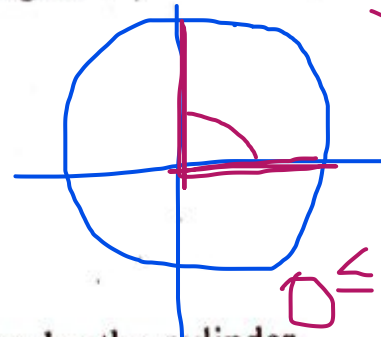
7. Find the area of the region D by setting double integral, where D is bounded by the parabola $y = x^2 - 2$ and the line $y = x$.

8. Write an equivalent integral with the order of integration reversed and then compute the integral :

$$\int_0^4 \int_0^{4-x} xy \, dy \, dx.$$

9. Calculate the Jacobian of transformation from rectangular to polar coordinates and hence evaluate the integral :

$$\int_0^2 \int_0^{\sqrt{4-y^2}} \frac{1}{\sqrt{9-x^2-y^2}} \, dx \, dy.$$



10. Find the volume V of the solid bounded above by the cylinder $y^2 + z = 4$ and below by $x^2 + 3y^2 = z$.

11. Evaluate the integral below, where D is the region bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$:

$$\iiint_D z \, dx \, dy \, dz.$$

12.

Let D be the region in the xy -plane that is bounded by the co-ordinate axes and the line $x + y = 1$. Use the suitable change of variable to compute the integral :

$$\iint_D \left(\frac{x-y}{x+y} \right)^6 dy dx.$$

Section III

13. State Green's theorem for simply connected regions. Use Green's theorem to find the work done by the force field $F(x, y) = (e^x - y^3)i + (\cos y + x^3)j$ along the circle $x^2 + y^2 = 1$ in anticlockwise direction.

14. Give the geometrical interpretation of the surface integral $\iint ds$ over piecewise smooth surface S . Evaluate the surface integral $\iint xz ds$ over the surface S which is the part of the plane $x + y + z = 1$ that lies in the first octant.

15. Verify Stokes' theorem for the vector field $F(x, y, z) = 2zi + 3xj + 5yk$ taking surface σ to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation and C to be the positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of σ in the xy -plane.

16. State and prove Divergence theorem.

17. Verify that the vector field $\mathbf{F}(x, y) = (e^x \sin y - y)\mathbf{i} + (e^x \cos y - x - 2)\mathbf{j}$ is conservative using cross partial test. Use a line integral to find the area enclosed by the ellipse :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

18. Let E be the solid unit cube with opposing corners at the origin and $(1, 1, 1)$ with faces parallel to co-ordinate planes. Let S be the boundary surface of E oriented with the outward pointing normal. If $\mathbf{F}(x, y, z) = 2xy\mathbf{i} + 3ye^z\mathbf{j} + x \sin z \mathbf{k}$, find the integral $\iint \mathbf{F} \cdot \mathbf{n} \, ds$ over surface S using divergence theorem.

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Roll No.

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No. of Question Paper : 92

Unique Paper Code : 32351303

I

Name of the Paper : C-7 Multivariate Calculus

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : III

Duration : 3 Hours

Maximum Marks : 75

Write your Roll No. on the top immediately on receipt of this question paper.)

All sections are compulsory.

All questions carry equal marks.

Section I

Attempt any six questions from this section.

1. Let f be the function defined by $f(x, y) = \frac{x^2 + 2y^2}{x^2 + y^2}$ for

$(x, y) \neq (0, 0)$.

(a) Find $\lim_{(x, y) \rightarrow (2, 1)} f(x, y)$.

(b) Prove that f has no limit at $(0, 0)$.

P.T.O.

2

The temperature at the point (x, y) on a given metal plate in the xy -plane is determined according to the form $T(x, y) = x^3 + 2xy^2 + y$ degrees. Compute the rate at which the temperature changes with distance if we start at $(2, 1)$ and move :

(a) parallel to the vector $\mathbf{j}$.

(b) parallel to the vector $\mathbf{i}$.

3. The Company sells two brands X and Y of a commercial soap, in thousand-pound units. If x units of brand X and y units of brand Y are sold, the unit price for brand X is $p(x) = 4,000 - 500x$ and for brand Y is $q(y) = 3,000 - 450y$.

(a) Find the total revenue R in terms of p and q .

(b) Suppose the brand X sells for \$ 500 per unit and brand Y sells for \$ 750 per unit. Estimate the change in total revenue if the unit prices are increased by \$ 20 for brand X and \$ 18 for brand Y.

4. $w = f\left(\frac{r-s}{s}\right),$

show that

$$r \frac{\partial w}{\partial r} + s \frac{\partial w}{\partial s} = 0.$$

5. Find the directional derivative of $f(x, y) = e^{x^2 y^2}$ at $P(1, -1)$ in the direction toward $Q(2, 3)$.

6. Find the absolute extrema of $f(x, y) = 2 \sin x + 5 \cos y$ in the rectangular region with vertices $(0, 0)$, $(2, 0)$, $(2, 5)$ and $(0, 5)$.

7. Let $\mathbf{R} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$ and $r = \|\mathbf{R}\|$, evaluate $\operatorname{div}\left(\frac{1}{r^3} \mathbf{R}\right)$.

Section II

Attempt any *five* questions from this section.

By using iterated integral, compute

$$\iint_R x \sqrt{1-x^2} e^{3y} dA,$$

where R is the rectangle $0 \leq x \leq 1$, $0 \leq y \leq 2$.

9. Evaluate the double integral :

$$\iint_D \frac{dA}{y^2 + 1},$$

where D is the triangular region bounded by $y = -x$ and $y = 2$. 14.

10. Evaluate the double integral

$$\int_0^2 \int_y^{\sqrt{8-y^2}} \frac{1}{\sqrt{1+x^2+y^2}} dx dy$$

by converting to polar co-ordinates.

11. Find the volume of the tetrahedron T bounded by the plane $2x + y + 3z = 6$ and the co-ordinates plane $x = 0$, $y = 0$ and $z = 0$.

12. Find the volume of the solid D bounded by the paraboloid $z = 1 - 4(x^2 + y^2)$ and the xy -plane.

13. Evaluate

$$\iint_D (x+y)^5 (x-y)^2 dy dx$$

by using change of variable $u = x + y$ and $v = x - y$ where D is the region in the xy -plane which is bounded by the co-ordinate axes and the line $x + y = 1$.

Section III

Attempt any *four* questions from this section.

14. Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{R},$$

where

$$\mathbf{F} = \frac{x}{\sqrt{x^2 + y^2}} \mathbf{i} - \frac{y}{\sqrt{x^2 + y^2}} \mathbf{j}$$

and C is the quarter circle path $x^2 + y^2 = a^2$, traversed from $(a, 0)$ to $(0, a)$.

15. Show that the vector field

$$\mathbf{F}(x, y, z) = \langle \sin z, -z \sin y, x \cos z + \cos y \rangle$$

is conservative and evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{R}$$

for any piecewise smooth path joining $A(1, 0, -1)$ to $B(0, -1, 1)$.

16. Use Green's theorem, to find the work done by the force field

$$\mathbf{F}(x, y) = (3y - 4x)\mathbf{i} + (4x - y)\mathbf{j}$$

when an object moves once counterclockwise around the ellipse $4x^2 + y^2 = 4$.

17. Use Stokes' theorem, to evaluate the line integral

$$\oint_C (3y \, dx + 2z \, dy - 5x \, dz)$$

where C is the intersection of the xy -plane and the hemisphere

$$z = \sqrt{1 - x^2 - y^2},$$

traversed counterclockwise as viewed from above.

18. Evaluate

$$\iint_S (\mathbf{F} \cdot \mathbf{N}) \, dS,$$

where $\mathbf{F} = x^2\mathbf{i} + xy\mathbf{j} + x^3y^3\mathbf{k}$ and S is the surface of the tetrahedron bounded by the plane $x + y + z = 1$ and the coordinate planes, with outward unit normal vector $\mathbf{N}$.